

Final Presentation

Building a Sales Forecasting Model



Meet the Team!



Apoorva Goel



Ning Wei



Harshal Gajera



Kuan-Hung Liu



Atharva Duragkar

Project Walkthrough



Introduction
Understand LoBs and
Sub-LoBs



Analyzing Sales Data
Developing an intuition
about the seasonality
and trend in each LoB



Data Processing
Exponential Smoothing
and removing the effect
of COVID-19 on sales



Forecasting models
Building a CNN-based
predictive model



**Results & Model
Deployment**
Discussing model
performance and
deployment



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Introduction

Understanding LoBs and Sub-LoBs

Who We Are



Mission

Enhancing personal and economic independence of people who are blind through employment opportunities



Vision

Blindness is not a barrier to employment

Commodities

Wide range of office supplies, hardware, and other goods.

Military Resales

Products tailored to meet the needs of U.S. military installations.

Services

Business solutions like call centers, logistics, and IT support.

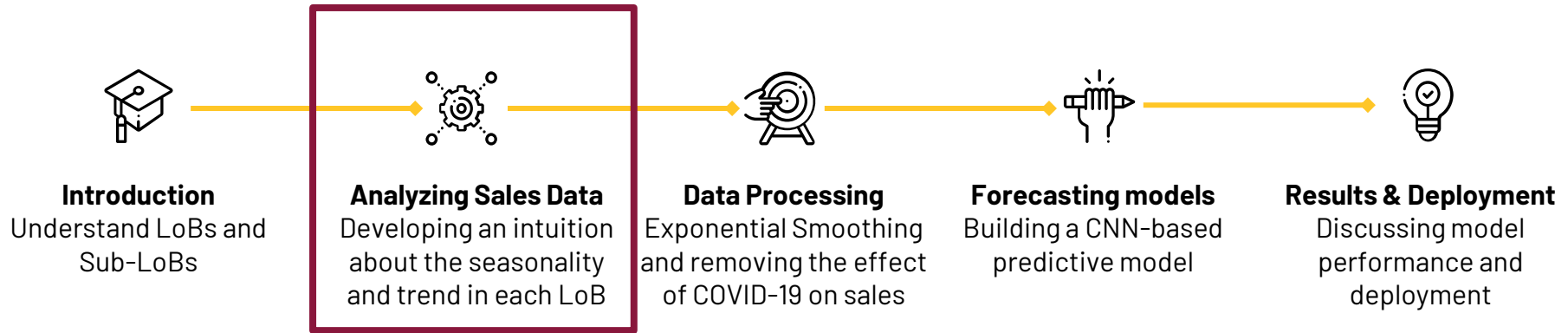
Textile

High-quality textiles, including uniforms and other fabric-based products.

Niche

Specialized products for unique markets, including custom manufacturing.





Analyzing Sales Data

Exploring Seasonality and Trend

Visual Inspection

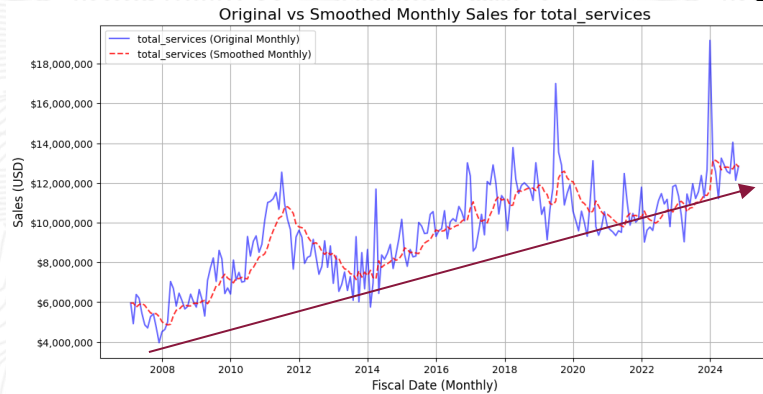
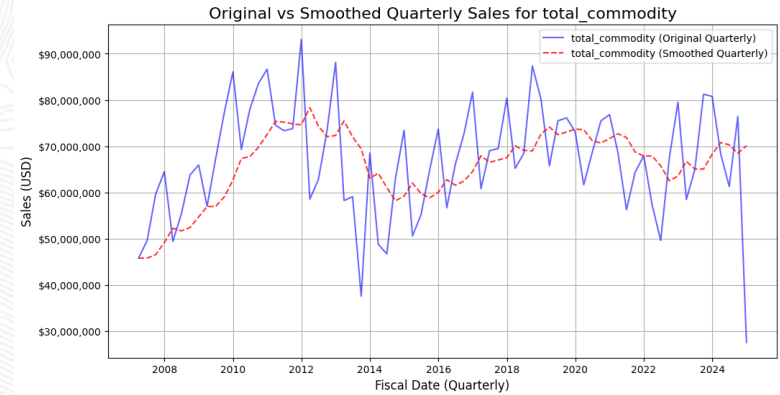
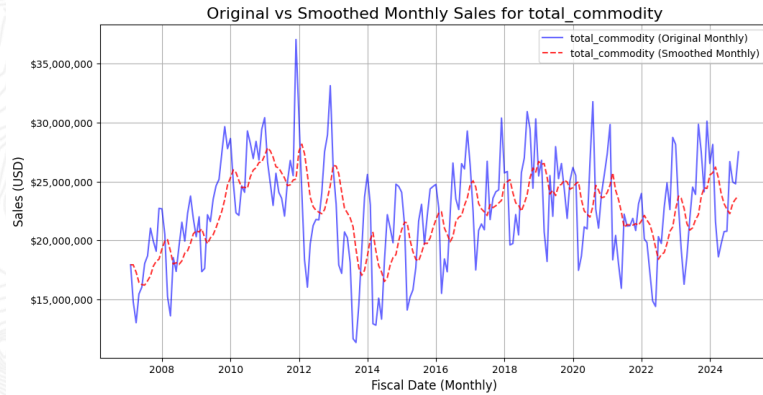
We start by plotting the time series for each line of business (or sub-lines) to visually inspect for any apparent trends or seasonality.

- **Trend:** Look for upward or downward long-term movements in the data.
- **Seasonality:** Look for repeating patterns at regular intervals (e.g., monthly, quarterly, yearly)

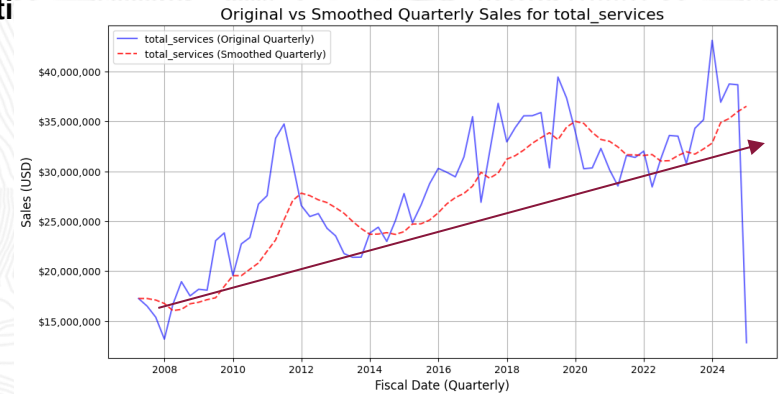
But looks can be deceiving which is why we need statistical methods to prove seasonality

Visual Inspection

- *Commodities: No clear trend, but a weak seasonality
- *Services: An upward trend, no visible seasonal pattern



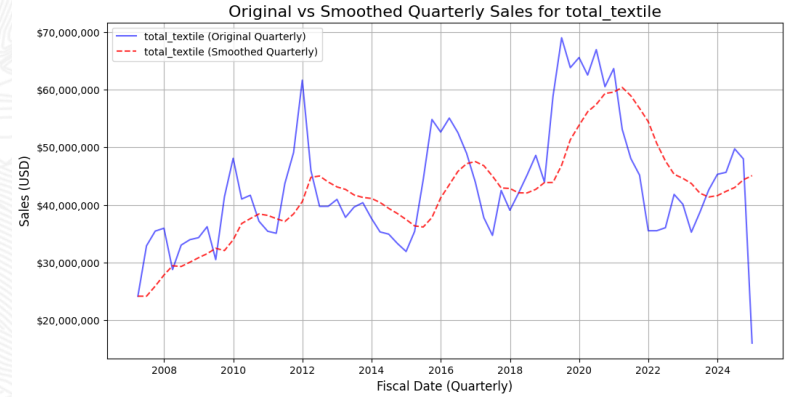
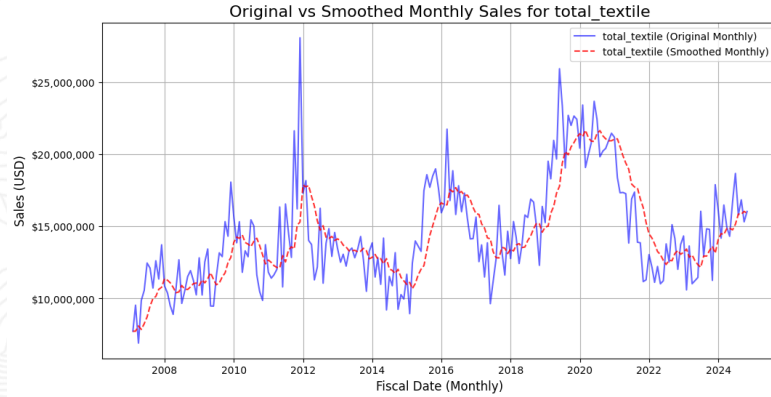
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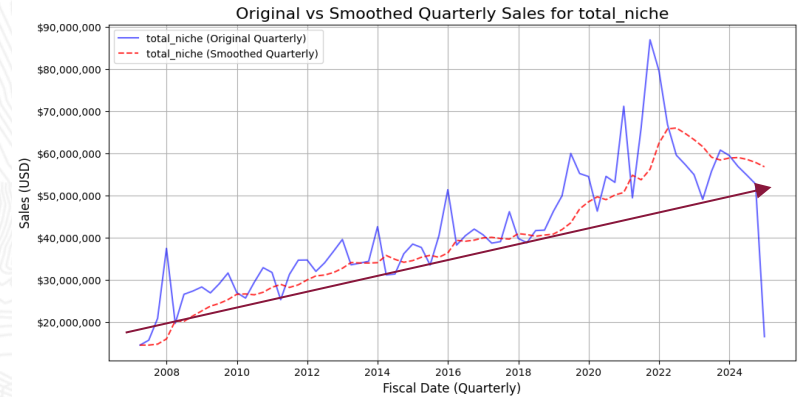
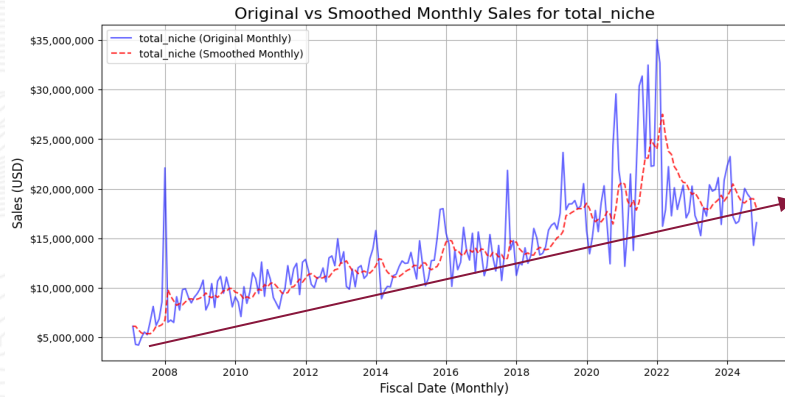
Services

Visual Inspection

- *Textile: No Clear Trend, Regular Seasonal Pattern
- *Niche: An upward trend, No visual seasonality noticed



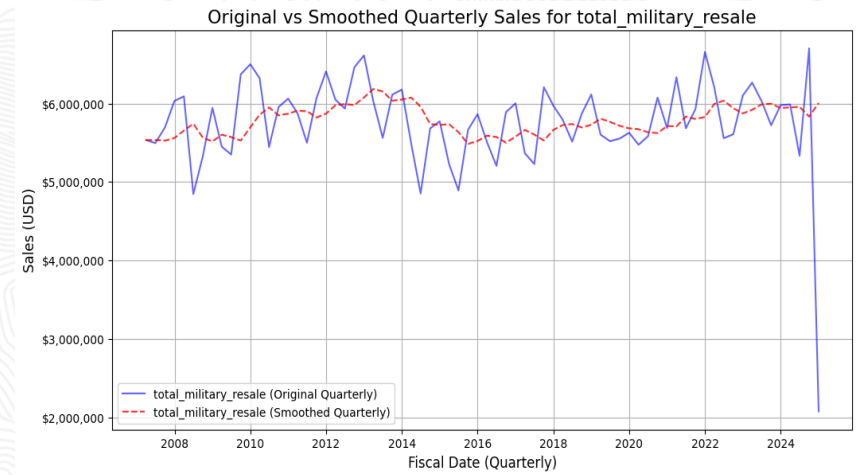
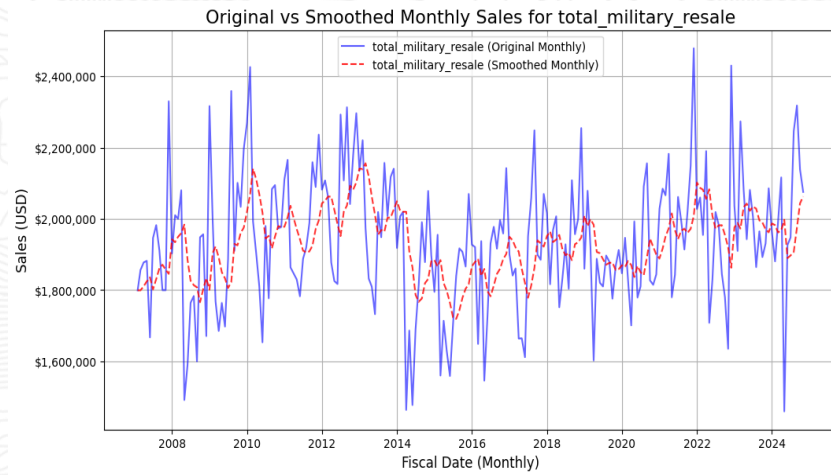
Textile



Niche

Visual Inspection

*Military Resale: No clear seasonality or trend

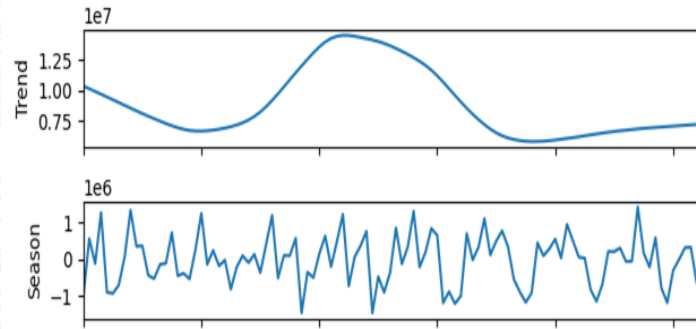


Military Resale

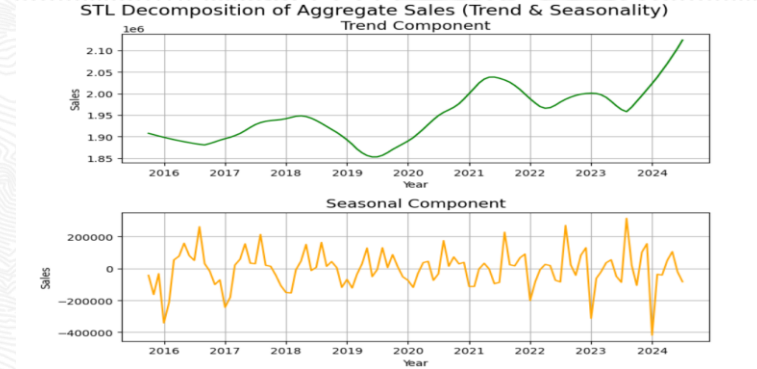
STL Decomposition

A versatile and robust method for breaking down a time series

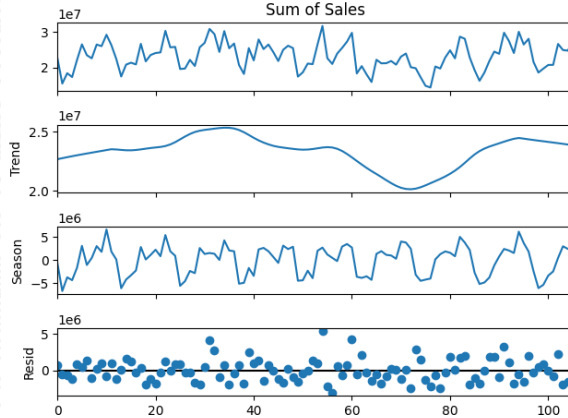
- It can handle any type of seasonality, not only monthly and quarterly data
- The seasonal component is allowed to change over time, and the rate of change can be controlled by the user
- The smoothness of the trend-cycle can also be controlled by the user
- It can be robust to outliers (i.e., the user can specify a robust decomposition), so that occasional unusual observations will not affect the estimates of the trend-cycle and seasonal components



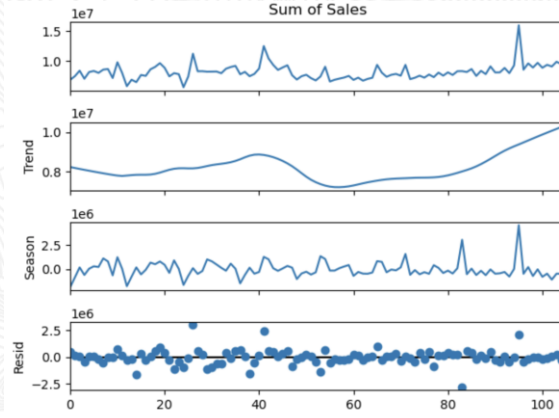
Textile



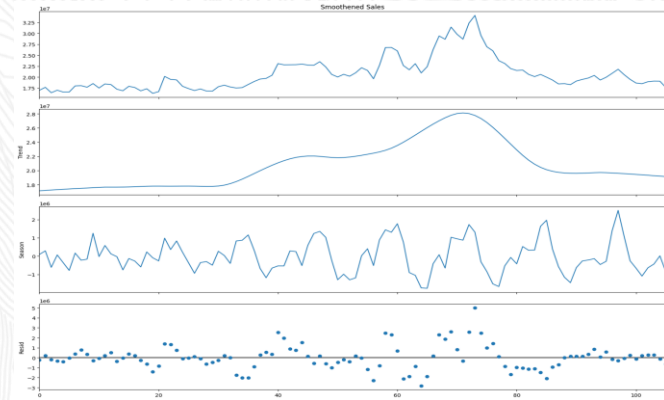
Military Resales



Commodities



Services



Niche

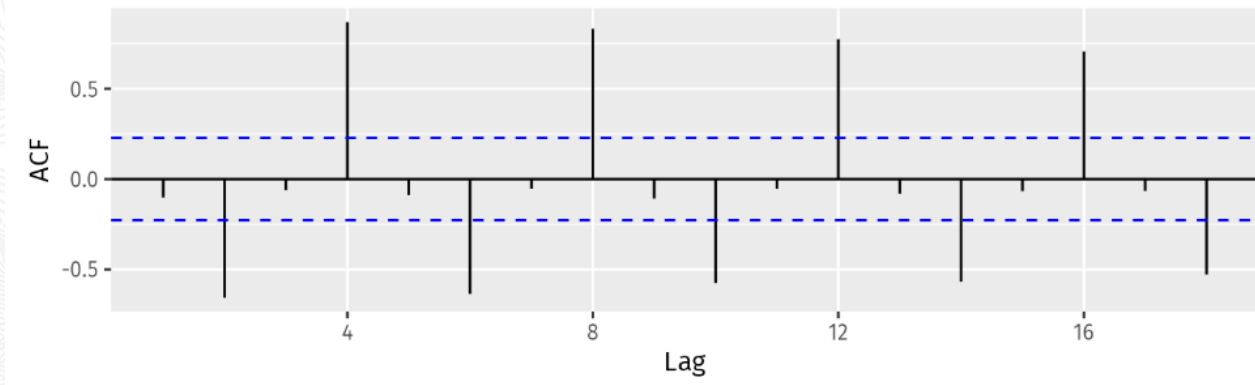
* STL Decomposition gives us the break down of the components, i.e., level, trend, seasonality and residuals.

Autocorrelation Function

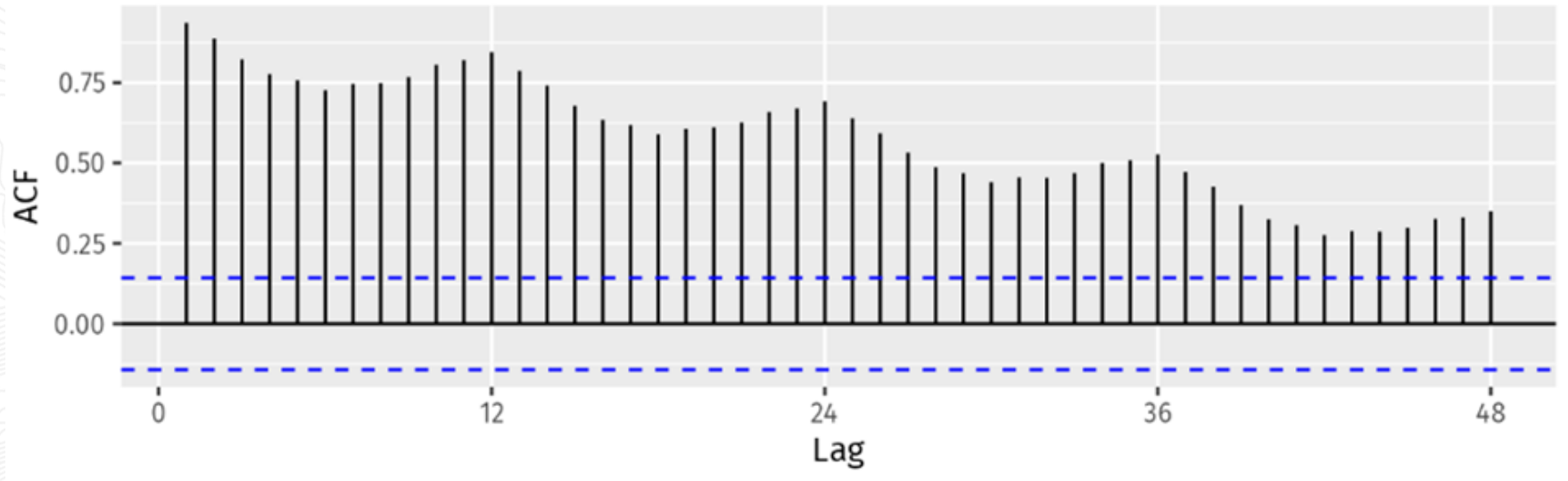
How well the time series at one point is related to its previous values at different lags

When the data has a **trend**, the autocorrelations for small lags tend to be large and positive because observations nearby in time are also nearby in size. So the ACF of trended time series tend to have **positive values that slowly decrease as the lags increase**.

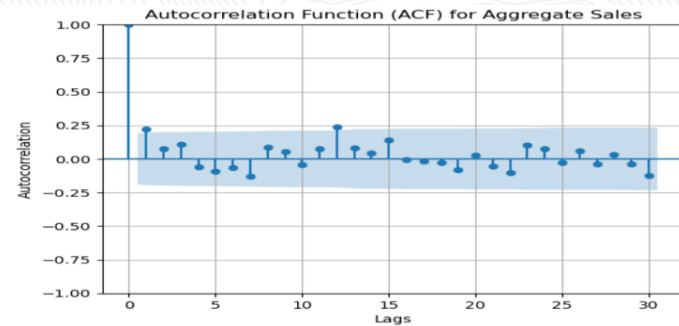
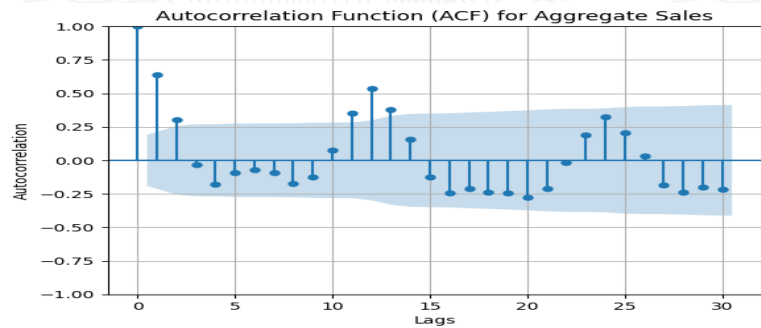
When the data is **seasonal**, the autocorrelations will be **larger for the seasonal lags** (at multiples of the seasonal frequency) than for other lags.



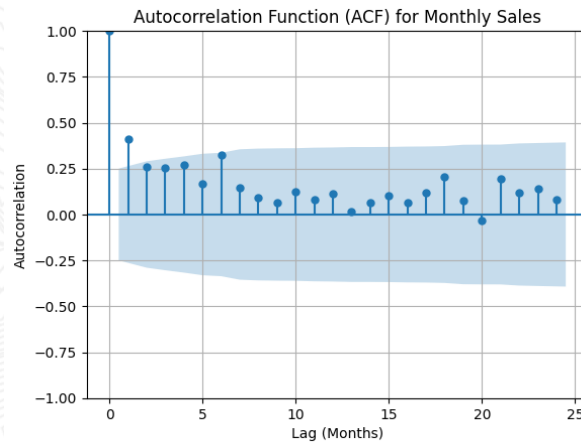
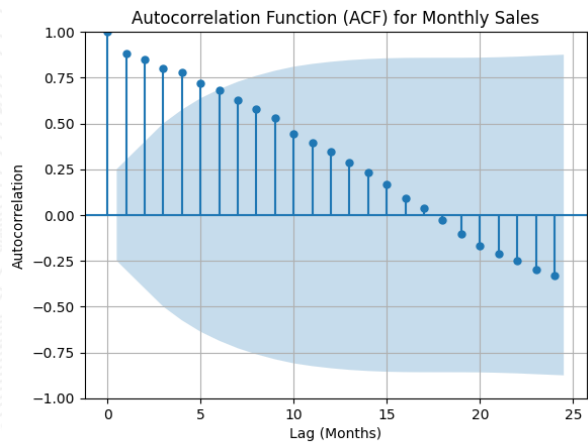
When the data have both trend and seasonal components, we see a combination of these effects.



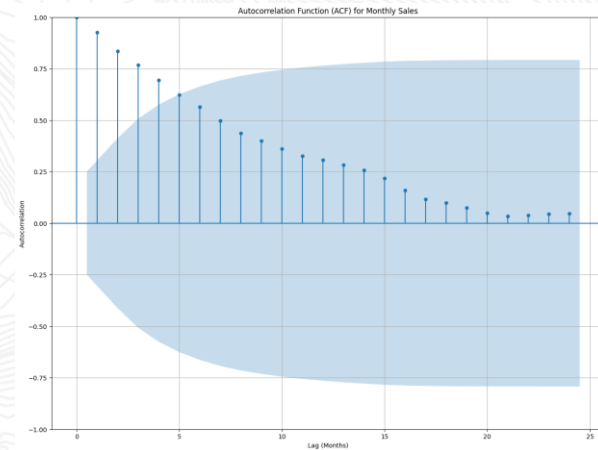
The slow decrease in the ACF as the lags increase is due to the trend, while the “scalped” shape is due to the seasonality.



Commodities



Military Resales

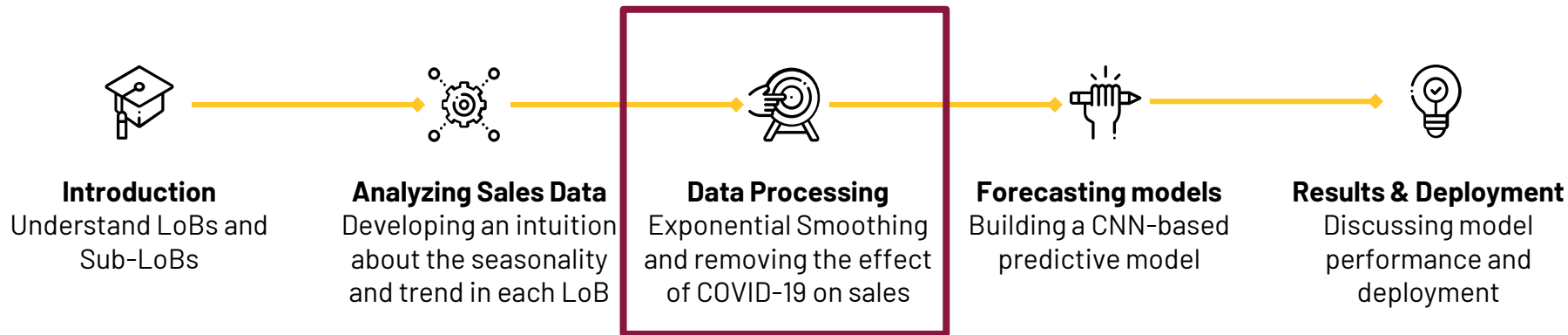


Textile

Services

Niche

* Commodities has a clear seasonality, whereas, textile and niche show a strong trend



Data Processing

Removing Outliers and Effect of Covid-19

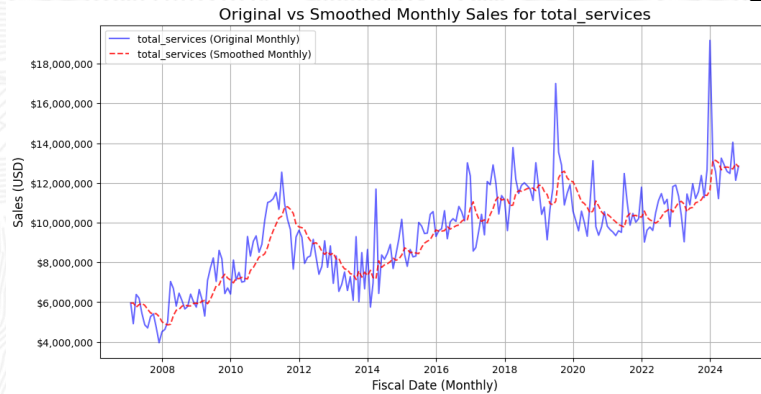
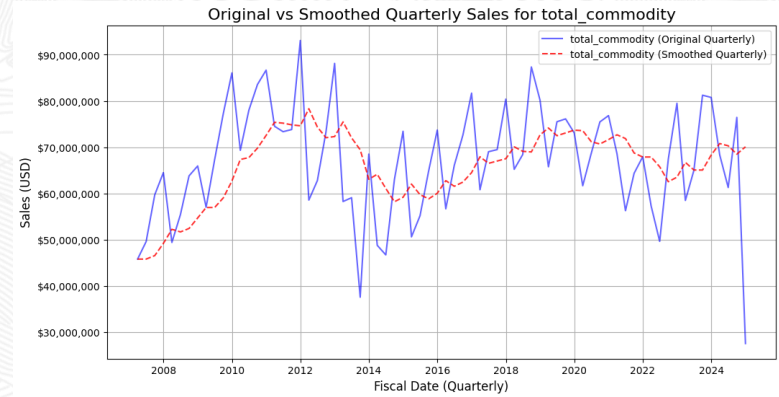
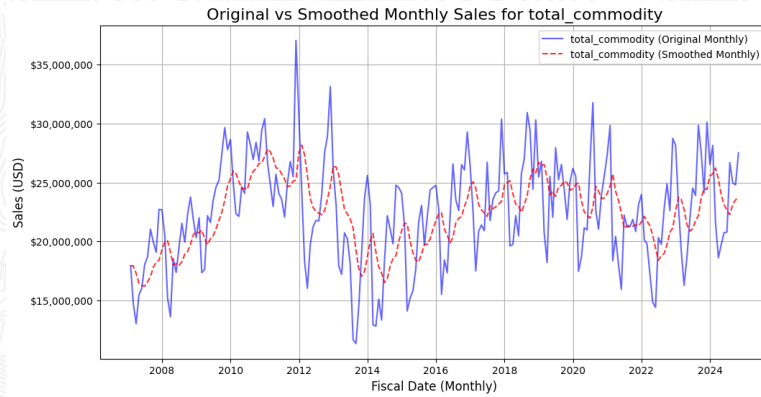
Fiscal Year Adjustment

Adjusting for Variability

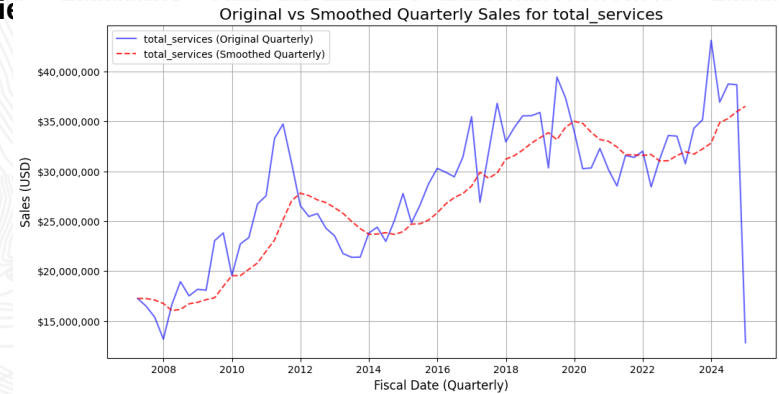
Exponential Smoothing

Adjusting for Variability

* The red dotted line shows the smoothed data



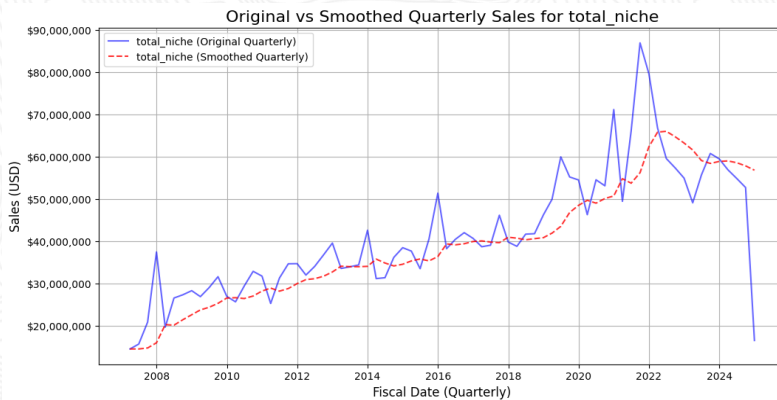
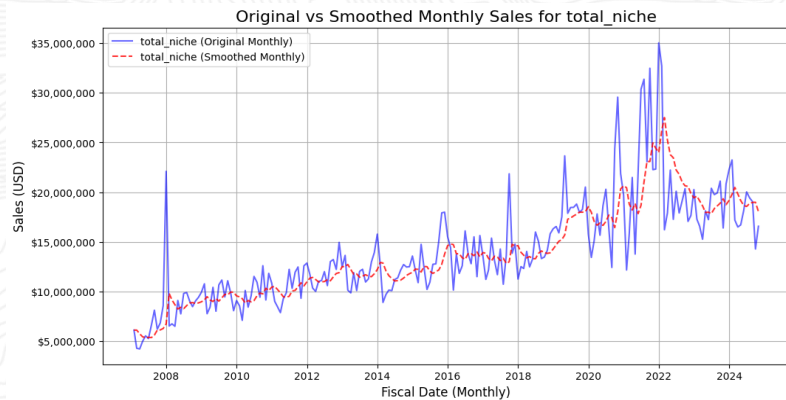
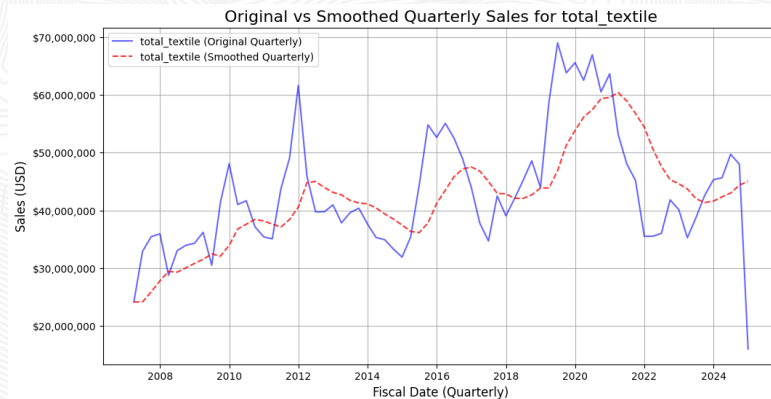
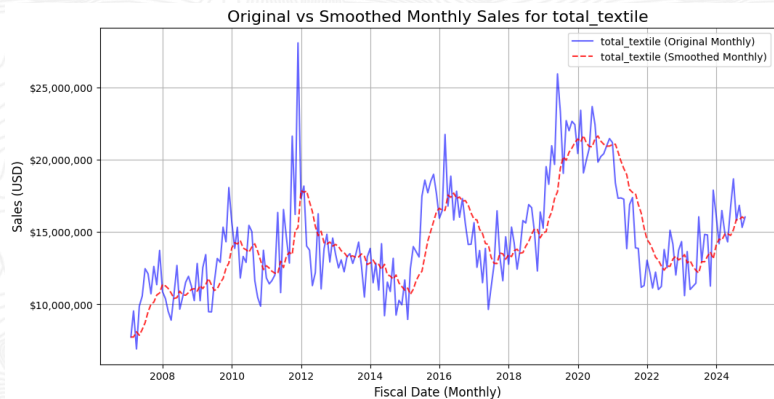
commodity



Services

Adjusting for Variability

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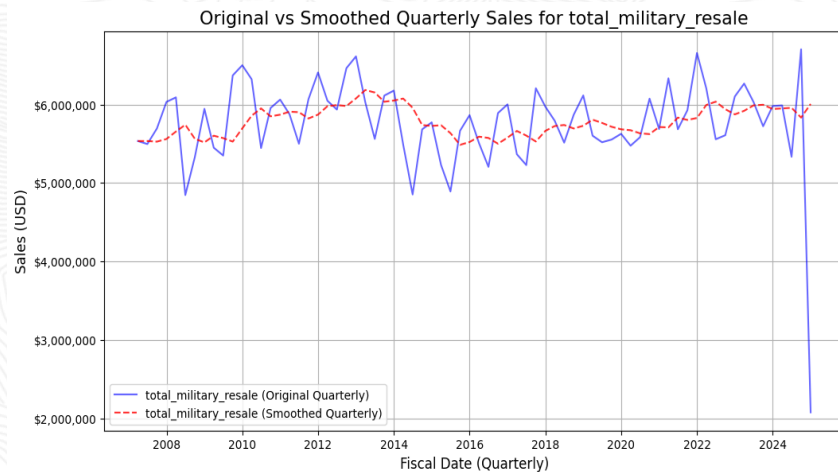
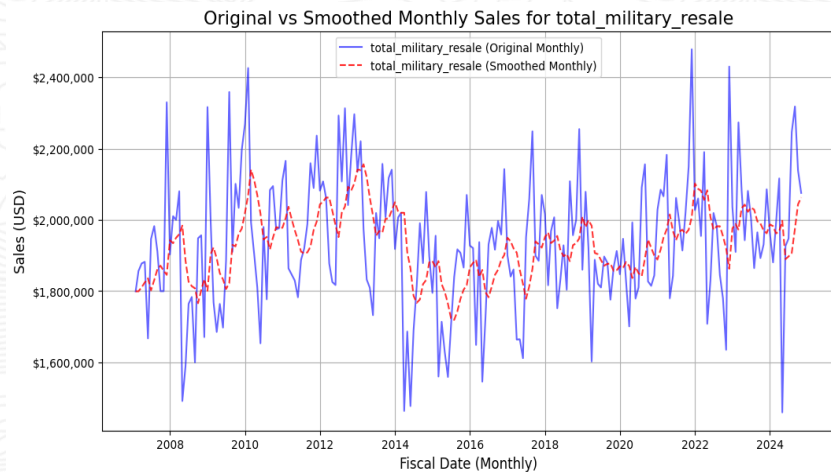


Textile

Niche

Adjusting for Variability

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Military Resale

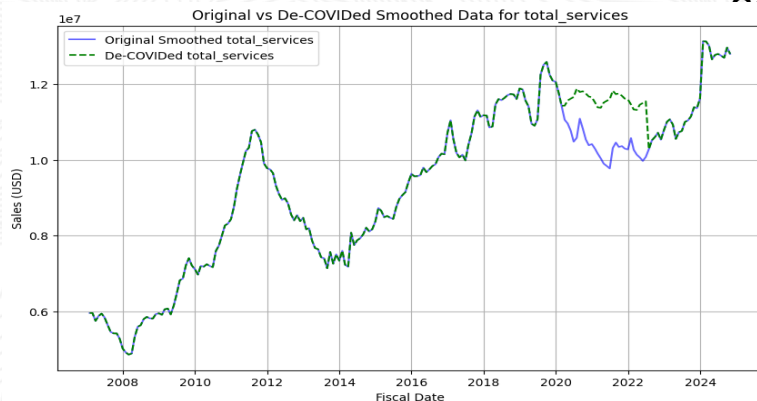
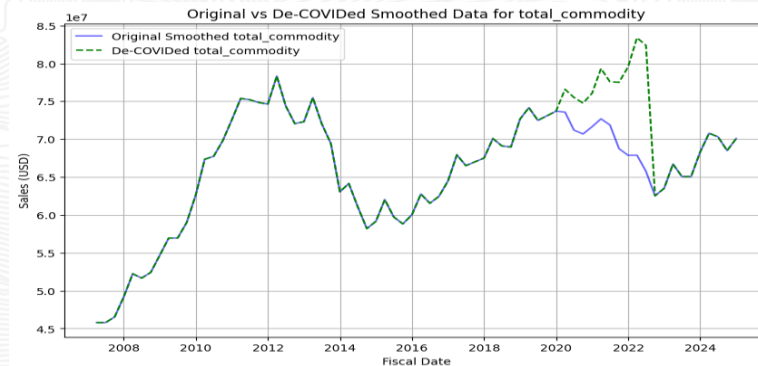
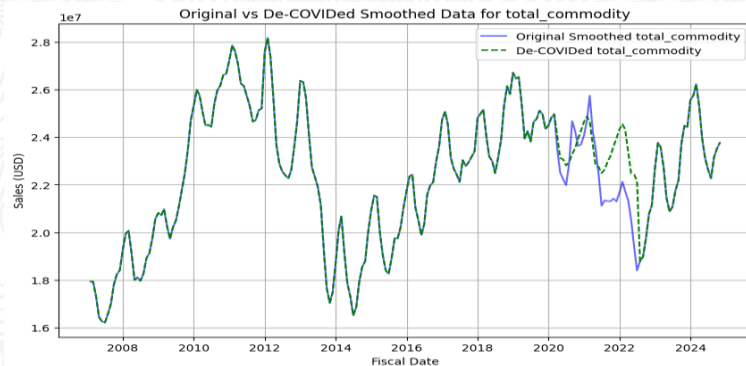
Covid-19 Adjustments

Holt's Winter Forecasting for June 2020 - June 2022

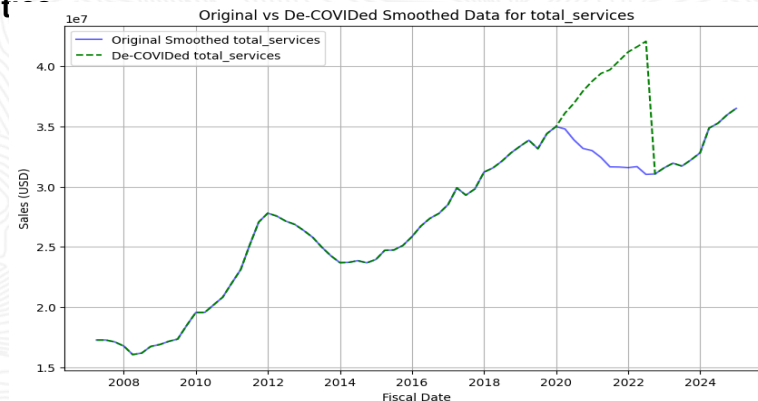
Our Approach

1. Split the fiscally adjusted dataset into three
 - Train Data : from Jan 2007 - May 2020
 - Test Data : from June 2020 - June 2022
 - Post-test Data : July 2022 - August 2024
1. Apply Holt-Winters Exponential Smoothing method on training data
2. Forecast for the period from June 2020 to June 2022 based on smoothed data
3. Apply Holt-Winters smoothing to the actual smoothed test data (June 2020 to June 2022)
4. Apply smoothing to the post-July 2022 smoothed data
5. Combine the smoothed and forecasted data into a final dataframe

Covid-19 Adjustments

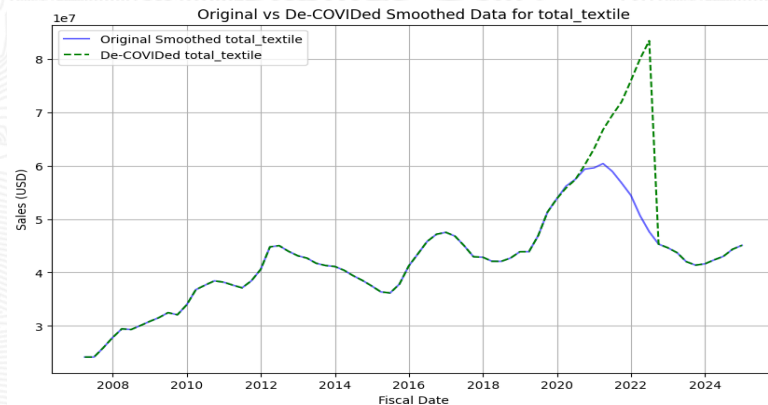
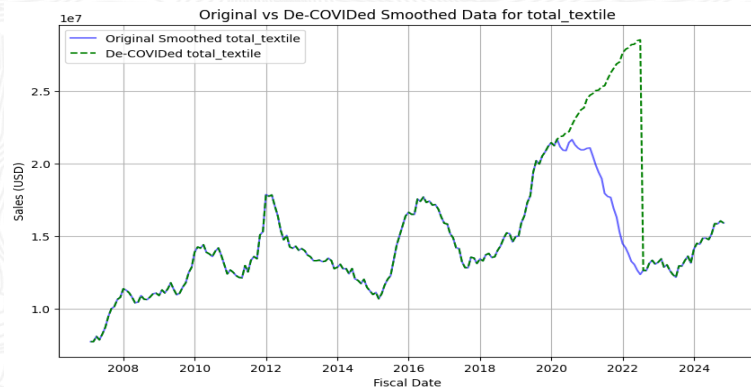


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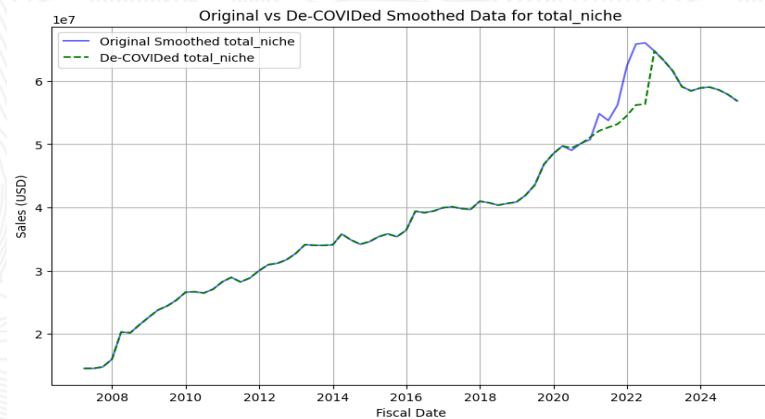
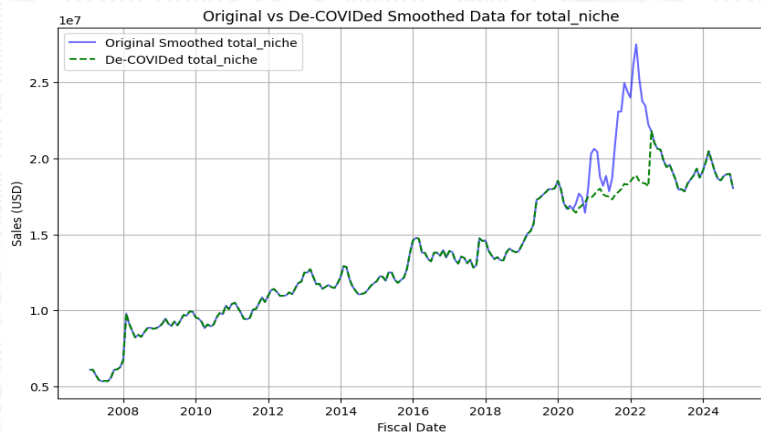


Services

Covid-19 Adjustments

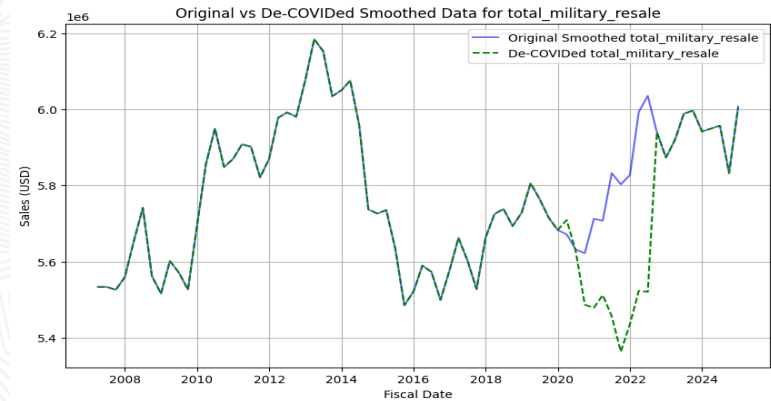
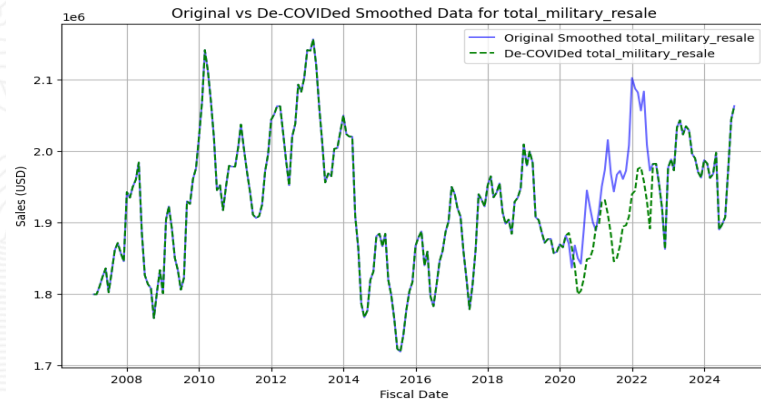


Textile

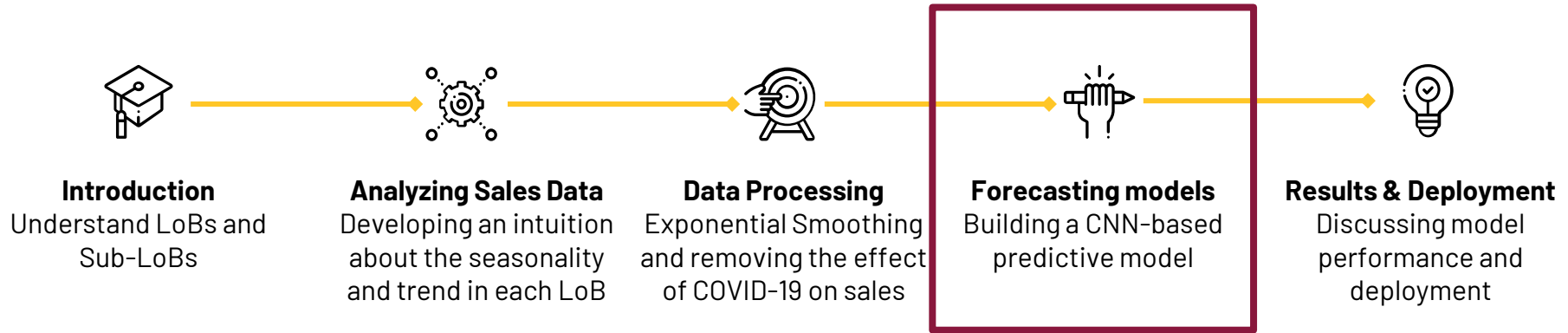


Niche

Covid-19 Adjustments



Military Resale



Forecasting Models

Forecasting Models: CNNs

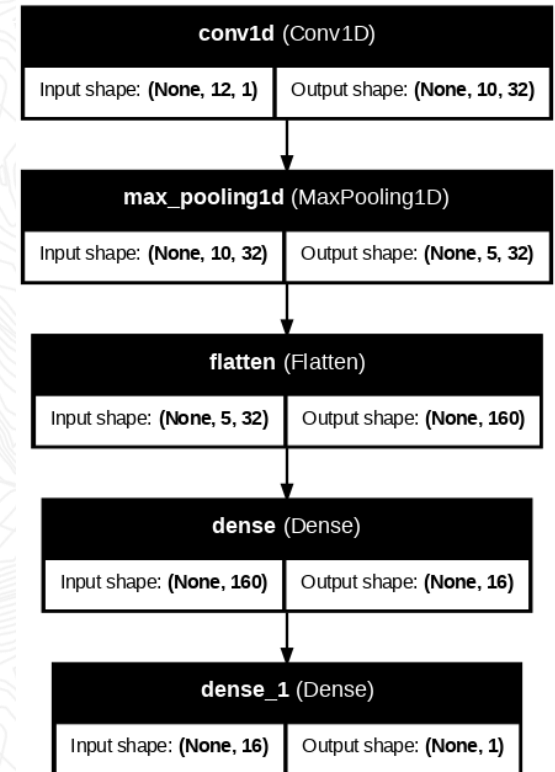
Overview: Deep learning model adapted for time series forecasting.

Features:

- Captures **nonlinear trends** and complex patterns.
- Handles **large, multivariate datasets** with automatic feature extraction.

Application:

- **Example:** Predicting sales spikes based on historical and external factors.





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Results & Deployment

Discussing model performance and deployment

Results and Deployment

CNN Model Performance

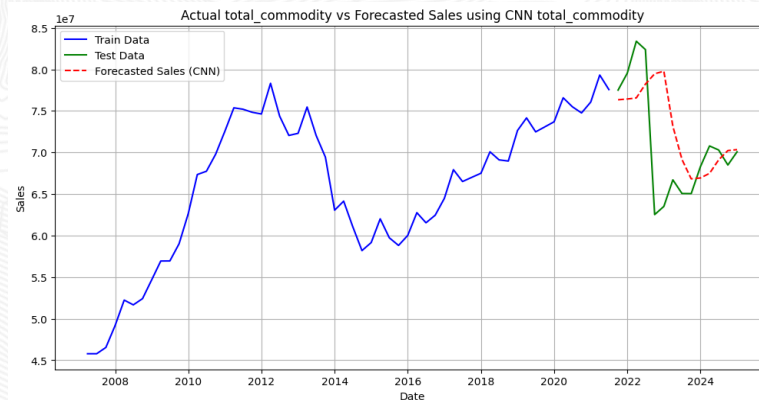
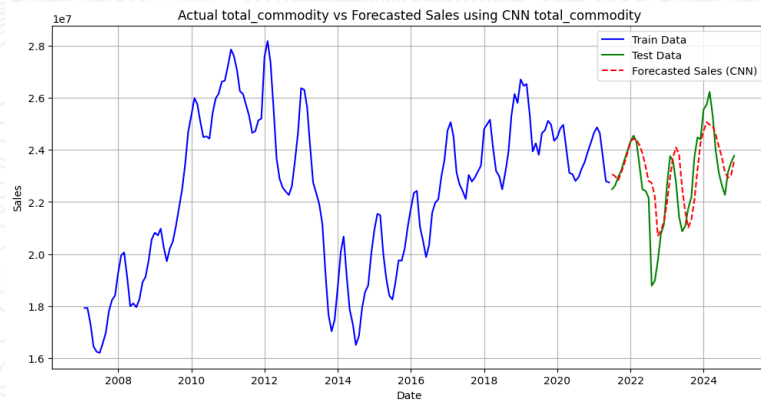
CNN Model result

MAPE of Each LoB Data by Monthly & Quarterly

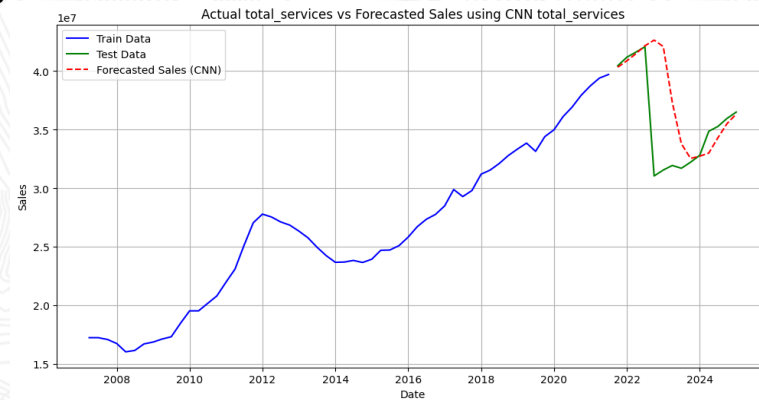
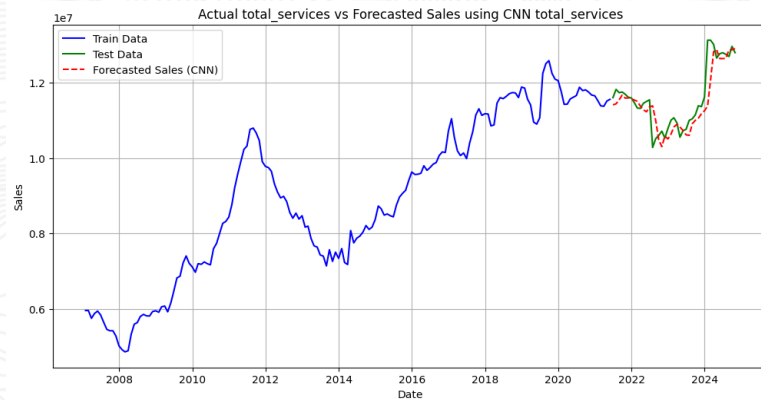
MAPE	Niche	Textile	Commodity	Service	Military
Monthly	3.82%	13.4%	3.73%	2.30%	1.61%
Quarterly	3.37%	20.01%	7.23%	7.62%	2.02%

*In most of situation, Monthly data have better performance than Quarterly data

CNN Model result

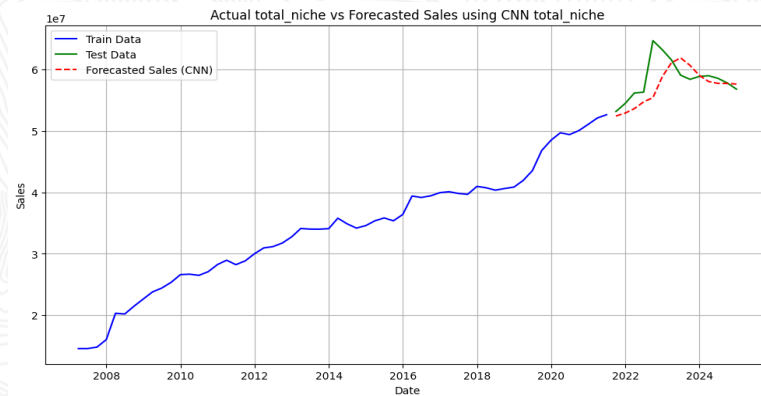
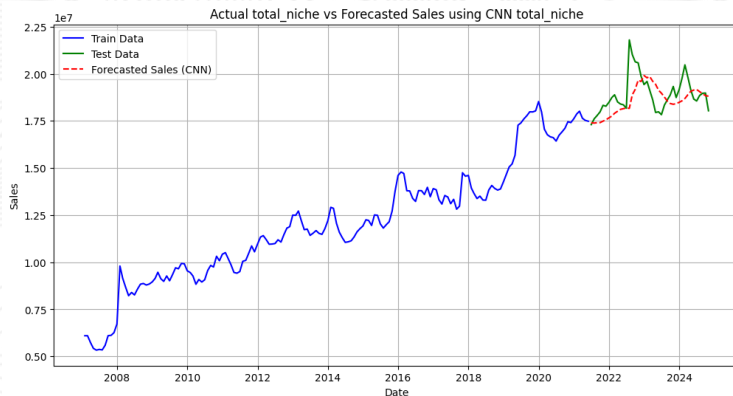
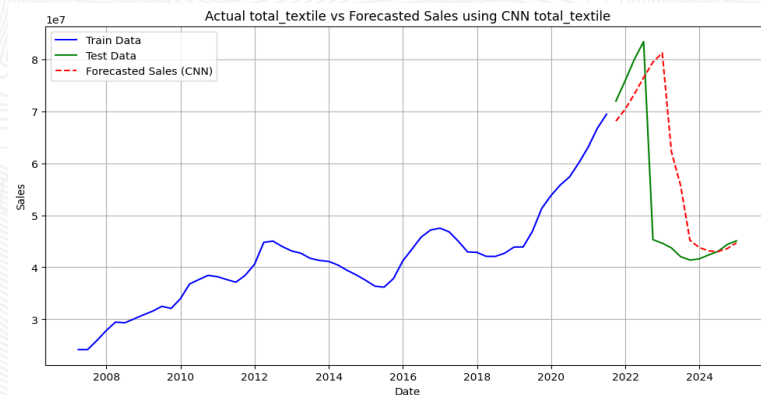
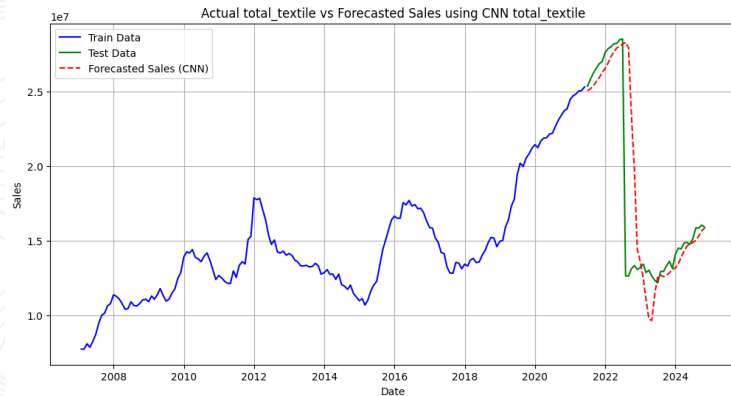


Commodities



Services

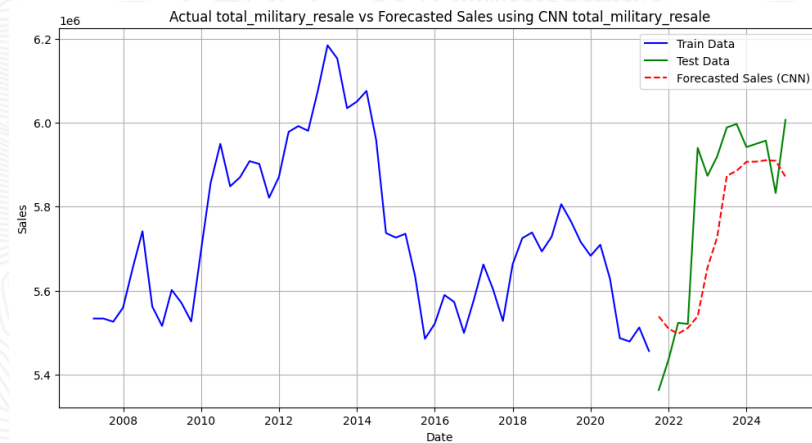
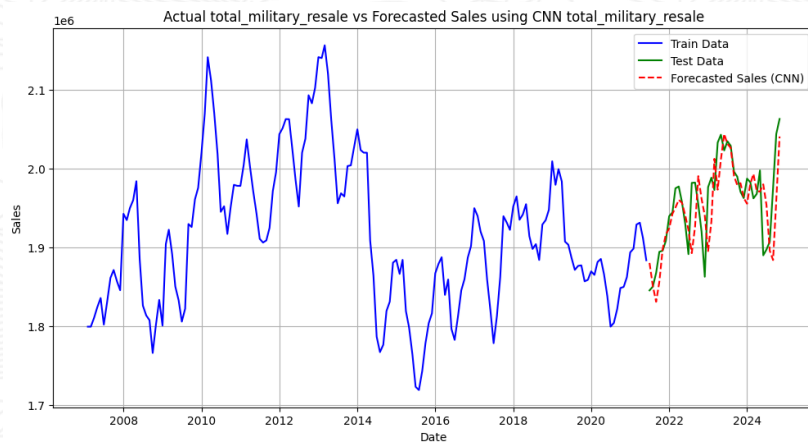
CNN Model result



Textile

Niche

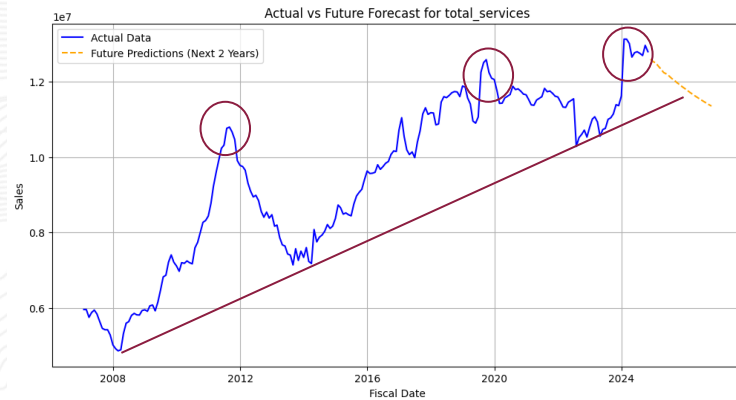
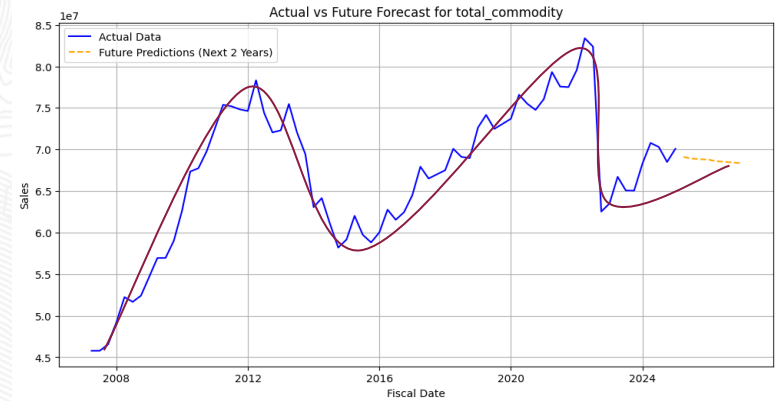
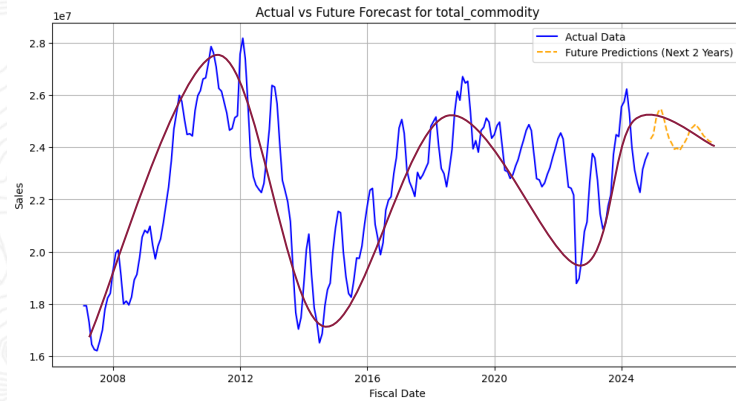
CNN Model result



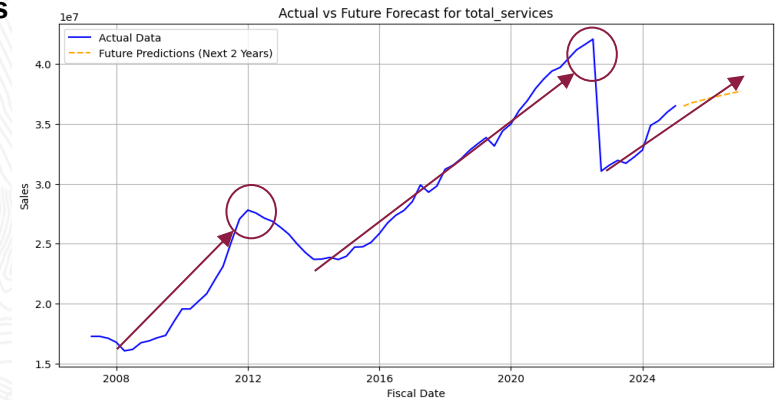
Military Resale

Forecasting for Next 24 Months

Forecasting

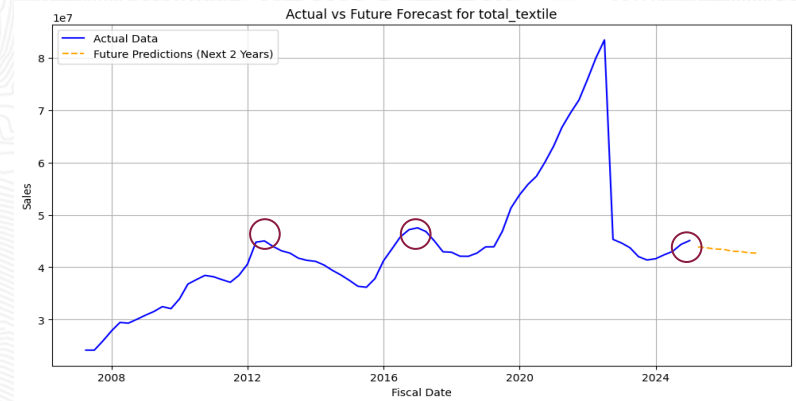
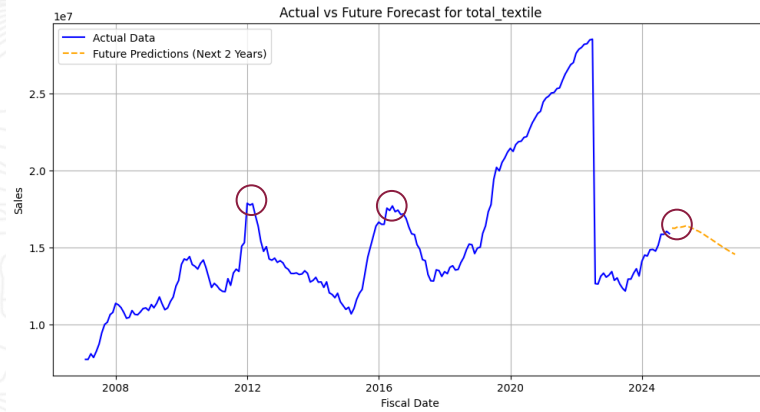


Commodities

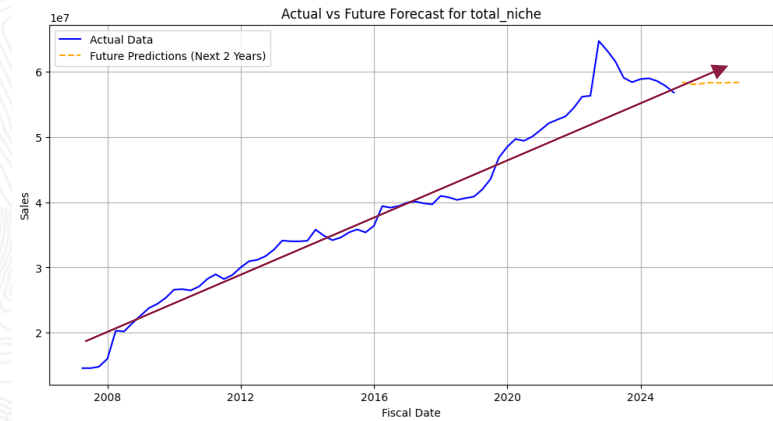
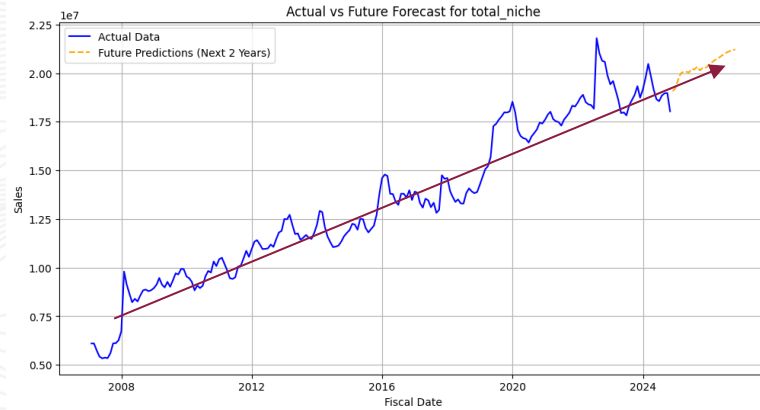


Services

Forecasting

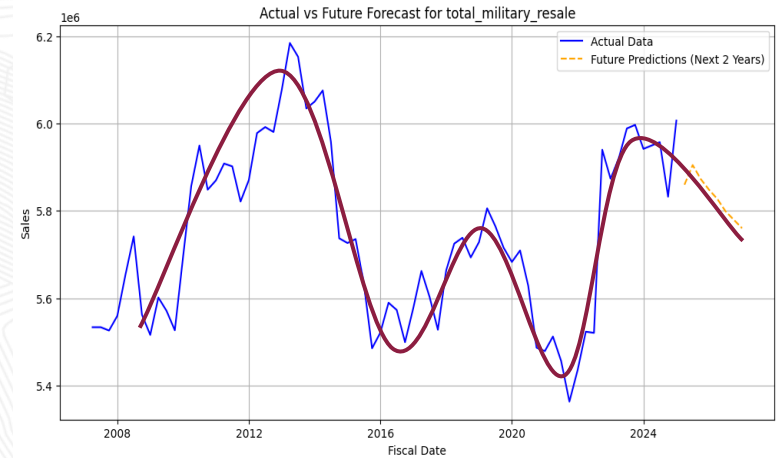
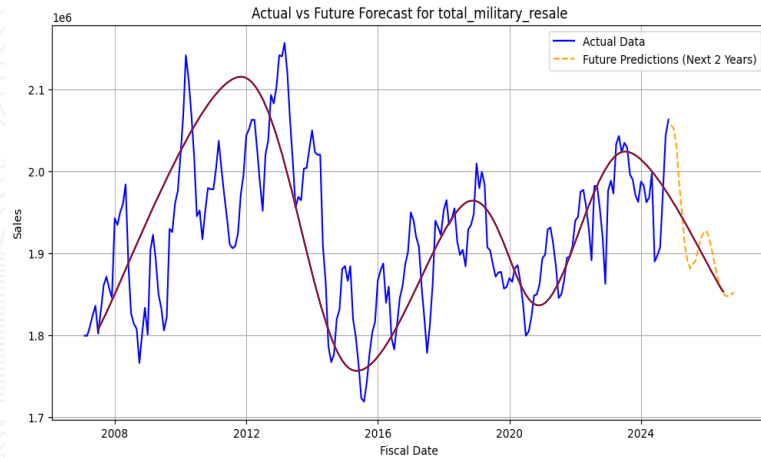


Textile



Niche

Forecasting

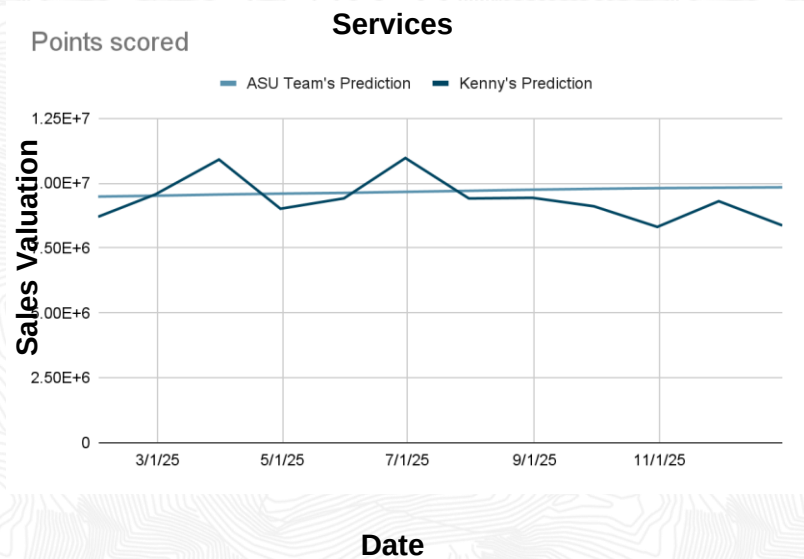
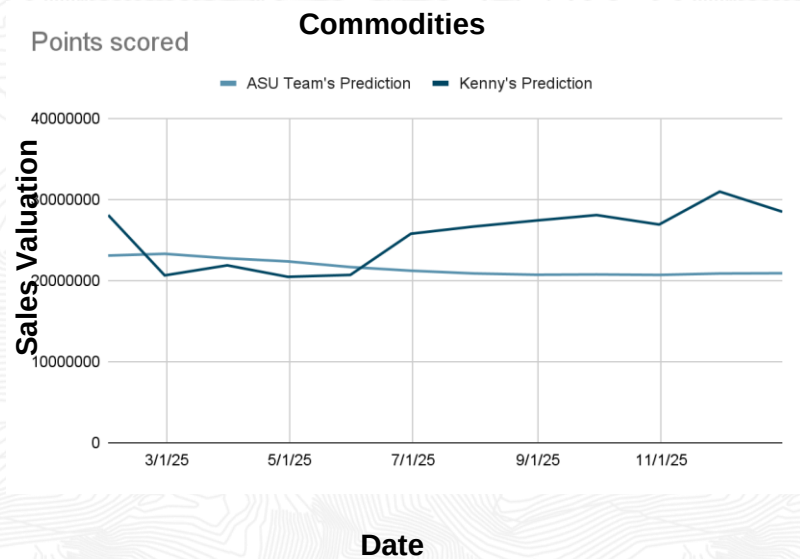


Military Resale

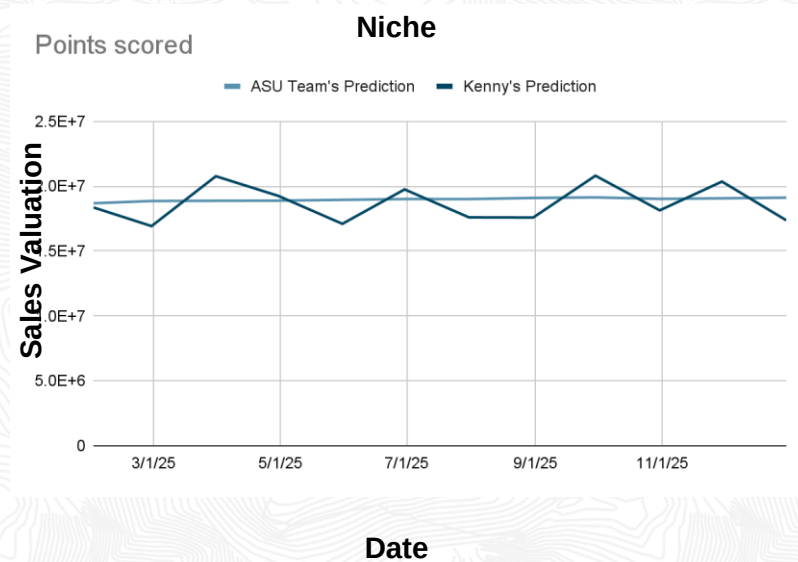
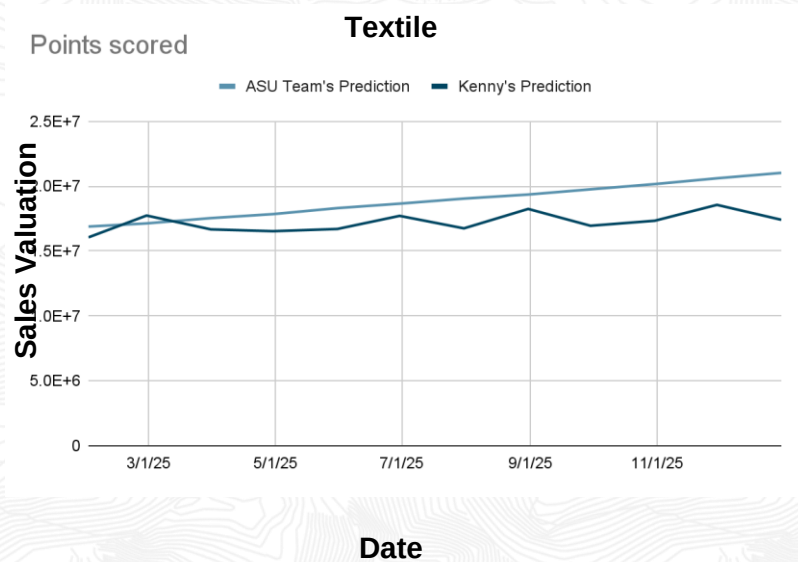
Forecast comparison

ASU team Vs Kenny

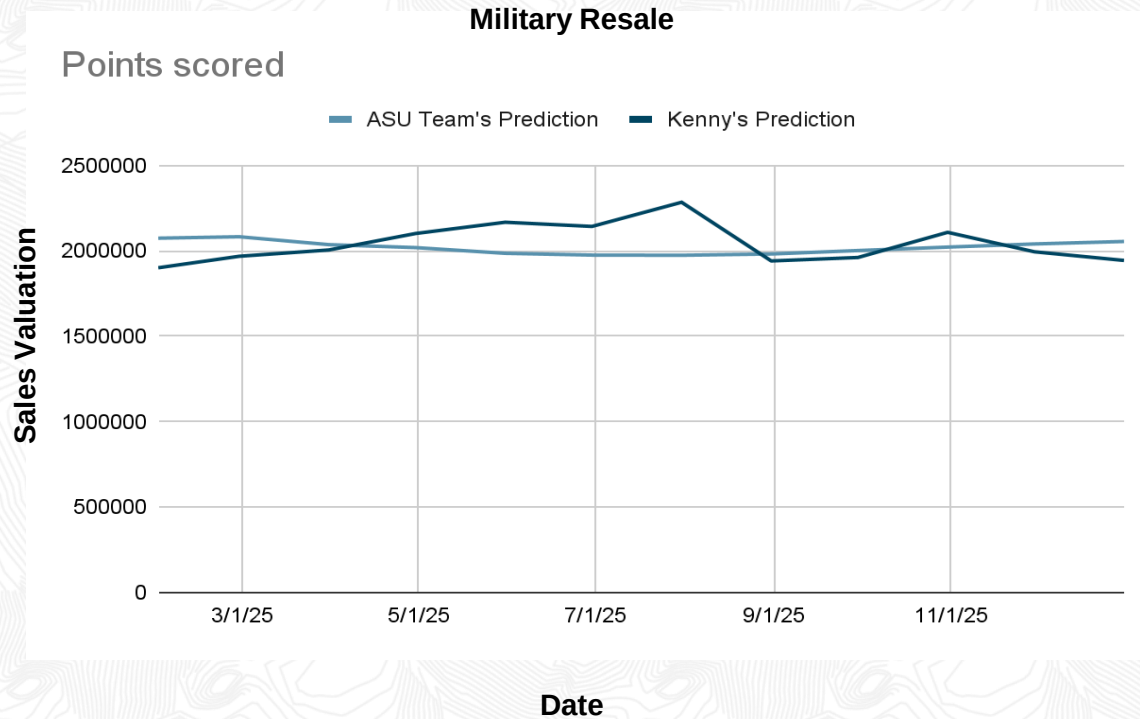
Forecast comparison



Forecast comparison



Forecast comparison



Model Deployment

Using Google Collaboration
Using an executable

Demo using Google Collaboration

Demo using the Executable

What is an “executable”?

It encapsulates the python code along with its dependencies into a single file so that it can run independently without the user having to download python on their system

Why use it?

- Python executables make our code **portable**, allowing it to run seamlessly on different platforms without users having to think about Python.
- Can be **shared easily**
- **User-friendly**



Questions?

THANK YOU!

